

ECG Teaching Note — Acute Pericarditis vs. STEMI

ECG interpretation guide · APP Fellowship Curriculum · 2026

1. Why This Distinction Matters

Acute pericarditis produces ST-segment elevation — and in a busy urgent care or emergency department, "ST elevation" triggers a cascade: the STEMI protocol, cath-lab activation, and anticoagulation. For a healthy 19-year-old with viral pericarditis, that cascade leads to unnecessary invasive angiography, anticoagulation risk, and a missed simple diagnosis. Conversely, a territorial STEMI dismissed as "probably pericarditis" in an older adult is a fatal error. The ECG is the first and fastest discriminator, and APP fellows must read the *pattern*, not just "ST elevation = bad."

2. The Four Stages of Pericarditis on ECG

Stage	Timing	Findings
I — Acute	First days	Diffuse concave (saddle-shaped) ST elevation in I, II, aVL, aVF, V2–V6; PR-segment depression in the same leads; reciprocal ST depression + PR elevation isolated to aVR . No pathologic Q waves.
II — Early repolarization phase	Days to ~2 weeks	ST segments return to baseline; T-wave amplitude decreases (flattening).
III — T-wave inversion	Weeks	T waves invert in the leads that previously showed ST elevation — can be mistaken for ischemic T-wave inversion.
IV — Resolution	Weeks to months	T waves normalize. Up to 10–15% may have persistent nonspecific changes.

PR-segment depression, present in up to 80% of stage I cases, is the single most specific ECG finding for pericarditis — caused by atrial epicardial inflammation (atrial repolarization vector shift).

3. Side-by-Side: Pericarditis vs. STEMI

Feature	Acute Pericarditis	STEMI
ST distribution	Diffuse — multiple vascular territories (I, II, aVL, aVF, V2–V6)	Territorial — one coronary artery's distribution (e.g., V1–V4, II/III/aVF)
ST morphology	Concave ("saddle-shaped"), upsloping	More often convex ("tombstone") or straightened
PR segment	Depressed (elevation in aVR)	Not depressed
Reciprocal ST depression	Only in aVR	Mirror-image territories (e.g., ST depression in I/aVL with inferior STEMI)
The aVR sign	ST elevation in aVR with diffuse ST elevation elsewhere → pericarditis	ST depression in aVR is normal in STEMI (reciprocal to widespread elevation)
Pain character	Sharp, pleuritic, positional — worse supine, better leaning forward; trapezius radiation	Pressure, heaviness, "sweaty and scared"; often with exertion; not positional
Typical patient	Young (17–40), healthy, recent URI	Older, risk factors (though not always)
Friction rub	May be present	Never
Troponin	Usually normal (elevated → myopericarditis)	Elevated with rise-and-fall pattern
Q waves	Never	May develop within hours
Evolution	Days to weeks (4 stages)	Hours (Q waves, T inversion within hours–days)

THE AVR SIGN — QUICK RULE

When you see diffuse ST elevation, look at **aVR**: **ST elevation in aVR + diffuse concave ST elevation + PR depression = pericarditis pattern**. A single-territory

convex ST elevation with reciprocal depression in a *different* territory (not just aVR) = STEMI pattern — activate. (One caution: ST elevation *in* aVR with diffuse ST *depression* elsewhere suggests left main or severe triple-vessel ischemia — an entirely different emergency.)

4. Pitfalls and Traps

- **Early repolarization (benign):** concave ST elevation in V4–V5, most often in young athletic men, but *no PR depression*, no fever, no friction rub, and unchanged on prior ECGs. Comparison with an old ECG is decisive.
- **Localized pericarditis:** pericarditis is occasionally focal (typically V3–V5) and can perfectly mimic anterior STEMI. Correlate clinically — history, rub, CRP, troponin, and echo; when in doubt, serial ECGs and cardiology consultation, not anticoagulation.
- **Myopericarditis:** pericarditis pattern + elevated troponin. Risk stratify with echo; some of these patients have reduced EF or arrhythmias and need cardiology involvement and true activity restriction.
- **Dressler syndrome:** weeks after MI or cardiac surgery — can look identical; get the history.
- **Spodick sign:** downsloping TP segment — suggestive of pericarditis, subtle; a bonus finding for advanced readers.

5. Decision Framework for the APP

1. **See diffuse ST elevation?** Run the pattern check: concave? PR depression? aVR sign? Pleuritic/positional pain? Recent URI? → pericarditis pathway.
2. **See a single territory of ST elevation?** Especially convex, with reciprocal depression, Q waves, or hemodynamic instability → STEMI pathway, activate.
3. **Ambiguous?** (focal changes, risk factors present, atypical story) → treat as ACS until excluded: serial ECGs, serial troponins, cardiology. Missing a STEMI is the error that cannot be undone.
4. **Diagnosed pericarditis?** → NSAID + colchicine, echo for everyone, look for tamponade physiology, restrict activity, and normalize CRP before tapering (see accompanying clinical reference).

Prepared as teaching source material for APP fellowship ECG curriculum.